Monthly Intelligence Report

Edition 001 — The Andalucía-Murcia Seismic Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

3,529

939 EHV · 2,590 HV

GRID LINES MAPPED

~22,765

~87,000 km coverage

MC SIMULATIONS

35.3 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 3,529 substations and ~22,765 grid lines spanning ~105,000 km.

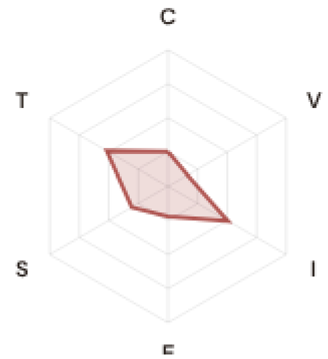
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Spain's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

SET Camp-redo

Tarragona, Cataluña · 110 kV

0.429	Medium	0.175	21th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.253 / 0.30 (84%)

I Infrastructure: 0.507 / 0.25 (203%)

S Saturation: 0.305 / 0.20 (153%)

V Voltage: 0.163 / 0.10 (163%)

E Economic: 0.220 / 0.10 (220%)

T Transition: 0.518 / 0.05 (1036%)

MODIFIERS

R3 Consequence: 1.022 ↑ amplifies

R4 Graph Criticality: 1.014 ↑ amplifies

R6a Restoration: 0.964 ↓ dampens

R6b Seismic: 1.247 ↑ amplifies

R7 Digital Readiness: 1.050 ↑ amplifies

This month's spotlight — automatically selected for April 2026 — is **SET Camp-redo** in Tarragona, Cataluña. With an R_final of 0.429 (Medium band, 21th fleet percentile), its risk profile is dominated by the T (Transition) component at 1036% of its weight ceiling, followed by E (Economic) at 220%. Modifiers collectively shift R_base by 30.9, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Spain's Plan de Digitalización / PNIEC 2030 planning requires.

CYBER HIGH EXPOSURE

633

17.9% of fleet

BLIND SPOTS

1630

High/Critical + R7 < median

AVG R7 MODIFIER

1.0079

Range [0.99, 1.05]

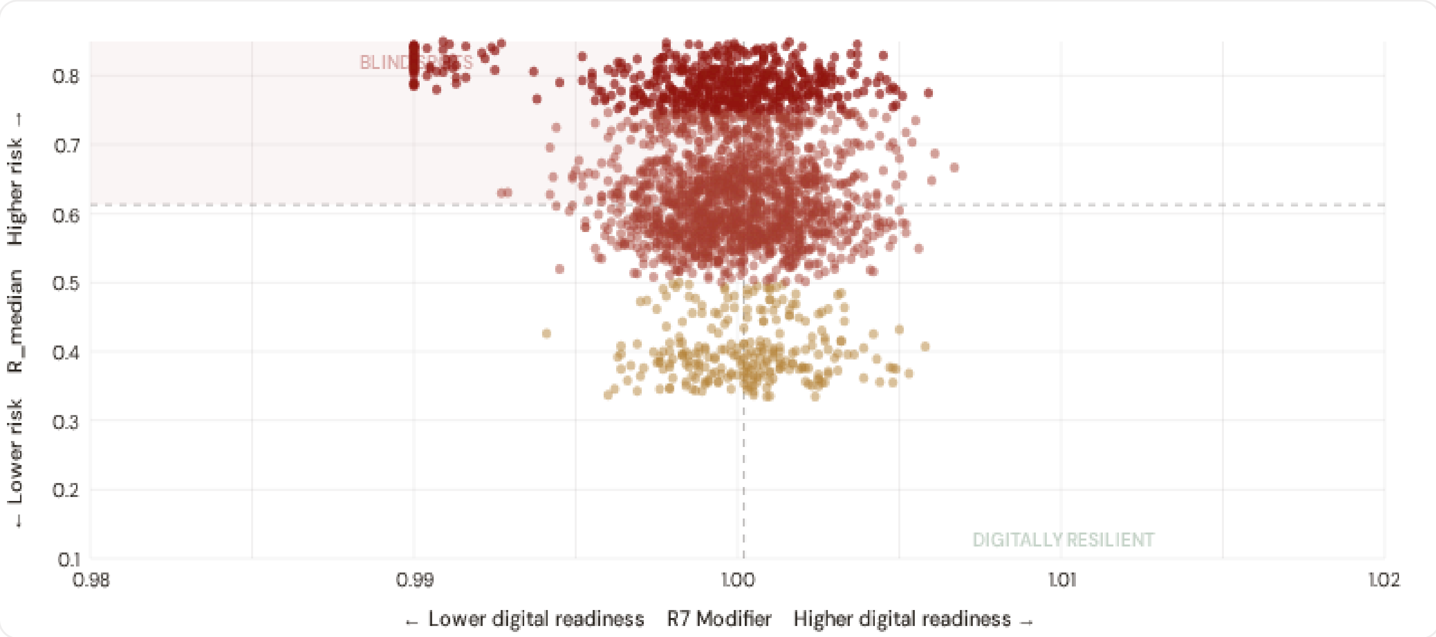
HIGH-CONSEQUENCE SUBS

1,652

46.8% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **1630 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0002). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Andalucía (345), Castilla-La Mancha (334), Castilla y León (269)**. Across the full fleet, 633 substations (17.9%) carry a HIGH cyber-exposure classification, while 2782 (78.8%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in major metropolitan areas (Madrid, Barcelona, Bilbao).

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €10–20/kWh

Heavy industry, petrochemical, large manufacturing — highest economic loss per interruption

1,522 substations (43.1%) High/Critical: **1444** (94.9%) Avg R: **0.718** Avg E: **0.311** / 0.10

Low 0 Med 78 High 885 Crit 559

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Diversified industry, logistics, food processing — significant continuity requirements

252 substations (7.1%) High/Critical: **131** (52.0%) Avg R: **0.523** Avg E: **0.332** / 0.10

Low 0 Med 121 High 119 Crit 12

🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

425 substations (12.0%) High/Critical: **303** (71.3%) Avg R: **0.574** Avg E: **0.407** / 0.10

Low 0 Med 122 High 275 Crit 28

🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

1,330 substations (37.7%) High/Critical: **666** (50.1%) Avg R: **0.525** Avg E: **0.350** / 0.10

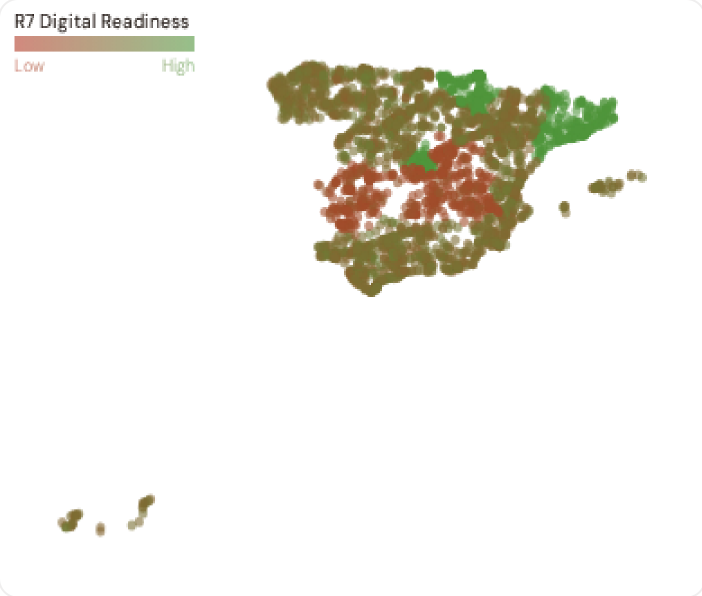
Low 153 Med 511 High 303 Crit 363

Value of Lost Load (VoLL) context: The ACER/CEPA 2022 pan-European VoLL study estimates Spain's sector-weighted average at approximately €6.4/kWh, with industrial customers at ~€11.8/kWh and residential at ~€3.5/kWh. The SSI's R3 consequence multiplier maps these sectoral weights to each substation's service territory. **1444 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Andalucía (606), Castilla-La Mancha (334), Extremadura (173)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of €10–20/kWh, compared to €1–3/kWh in rural agricultural zones. The fleet-wide average energy poverty rate is 10.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Provincia Digital Readiness Ranking

Provincias ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provincias combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Toledo ⚠ high R	0.9907
2	Guadalajara ⚠ high R	0.9908
3	Badajoz ⚠ high R	0.9908
4	Cuenca ⚠ high R	0.9908
5	Ciudad Real ⚠ high R	0.9909
6	Albacete ⚠ high R	0.9909
7	Cáceres ⚠ high R	0.9912
8	Almería ⚠ high R	0.9993
9	Cádiz ⚠ high R	0.9995
10	Jaén	0.9995
11	Castellón	0.9998
12	Santa Cruz de Tenerife ⚠ high R	0.9998
13	Las Palmas ⚠ high R	0.9998
14	León ⚠ high R	0.9998
15	Alicante	0.9999

Provincia-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Toledo** has the lowest average R7 (0.9907), while **Álava** leads at 1.0495 — a spread of 0.0588 across the R7 range. **21 provincias** combine below-median digital readiness with above-median grid risk, making them priority targets for España 2030 / Plan de Recuperación / PPE digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0079, median = 1.0002; 633 substations classified HIGH cyber-exposure; 1630 blind-spot substations (High/Critical risk + below-median R7); 756 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging provincias where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL

30

95 variables

4

High-frequency feeds

18

Regular refresh cycle

STATIC / DERIVED

8

Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
SEPE	SEPE (Servicio Público de Empleo Estatal)	MONTHLY	Provincia	1 var
IDAE	IDAE (Instituto para la Diversificación y Ahorro de la Energía)	QUARTERLY	Provincia	5 vars
CNMC	CNMC (Comisión Nacional de los Mercados y la Competencia)	ANNUAL	Provincia	8 vars
INE	INE (Instituto Nacional de Estadística)	ANNUAL	Provincia	8 vars
MITECO	MITECO (Ministerio para la Transición Ecológica)	ANNUAL	CCAA	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
CCNC	CCN-CERT (Centro Criptológico Nacional)	ANNUAL	National	2 vars
PROT	Protección Civil / DG Catástrofes	ANNUAL	Provincia	2 vars
CSN	CSN (Consejo de Seguridad Nuclear)	ANNUAL	Site-level	1 var
INE-E	INE Economic Statistics	ANNUAL	CCAA	3 vars
REE-S	REE Statistical Series	ANNUAL	CCAA	2 vars
ADIF	ADIF Rail Statistics	ANNUAL	Provincial	1 var
DGT	DGT (Dirección General de Tráfico)	ANNUAL	Provincial	1 var
CNMC-M	CNMC Monitoring Report	ANNUAL	DSO-level	2 vars
IGN	IGN/IGME (Instituto Geográfico Nacional)	STATIC	Provincia	3 vars
CDS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
NCSE	NCSE-02 Seismic Code	STATIC	Municipal	1 var
COG-ES	INE Municipal Registry	STATIC	Provincial	1 var
ISO9223	ISO 9223 Corrosion	DERIVED	Provincia	1 var

REE	REE (Red Eléctrica de España) — Transparency	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
AEMET	AEMET (Agencia Estatal de Meteorología)	HOURLY	Station	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (2 sources) — Grid load + procurement data

IDAE Registry · REE Transparency

Quarterly (1 sources) — DER registry + business data + macro indicators

DGT

Annual (17 sources) — Core grid quality + demographics + environment

CNMC · INE · MITECO · SEPE · IDAE · REE-S · CNMC-M · EEA · Eurostat · DESI · CCN-CERT · CSN · JRC · Protección Civil · ADIF · AEMET · INE-E

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · IGN · ISO 9223 · COG-ES/INE · NCSE-02

C.2 Fleet Score Snapshot

FLEET SIZE

3,529

substations scored

MEDIAN R

0.613

P5=0.254 · P95=0.950

HIGH + CRITICAL

2544

72.1% of fleet

HIGH CONFIDENCE

72%

2526 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 153 (4.3%) · Medium 832 (23.6%) · High 1582 (44.8%) · Critical 962 (27.3%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **3,529 substation scores remain unchanged** this edition. The fleet median R of 0.613 (mean 0.614) reflects a distribution skewed toward the lower bands, with 4.3% in Low and 23.6% in Medium. The first material score changes are expected when CNMC publishes the annual TIQE indicators (affecting C1–C4 and R6a) and when IDAE refreshes the Registro de Instalaciones de producción de energía eléctrica (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Provincia-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — R&D + economic structure	§6
L3	ENHANCED	V_socio Fiscal Enrichment — INE + MITECO + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — INE net migration	§5
L5	DATA	95/95 variables operational (100%). 30 data sources total.	§8, §12
G1	DATA	IDAE upgraded to quarterly registry — Provincia-level DER registry	§12
G2	DATA	IGN upgraded to live API — Provincia-level seismic data	§12
G3	DATA	OSM upgraded to Overpass API — 3,529 real substations	§12
G4	NEW	R6b Seismic Hazard modifier — NCSE-02 classification from IGN	§5
G5	NEW	Network & Topology layer (H): 7 variables — REE Grid Plan, seismic analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Andalucía-Murcia Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Andalucía-Murcia seismic corridor · **002** DER saturation hotspots (Valencia/Murcia) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Nuclear fleet dependency and phase-out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Provincia Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Andalucía-Murcia, Why Now

ANDALUCÍA-MURCIA SUBSTATIONS

688

294 EHV · 394 HV

HIGH BAND

637 + 51

100.0% of corridor

CORRIDOR MEDIAN R

0.628

Fleet median: 0.613

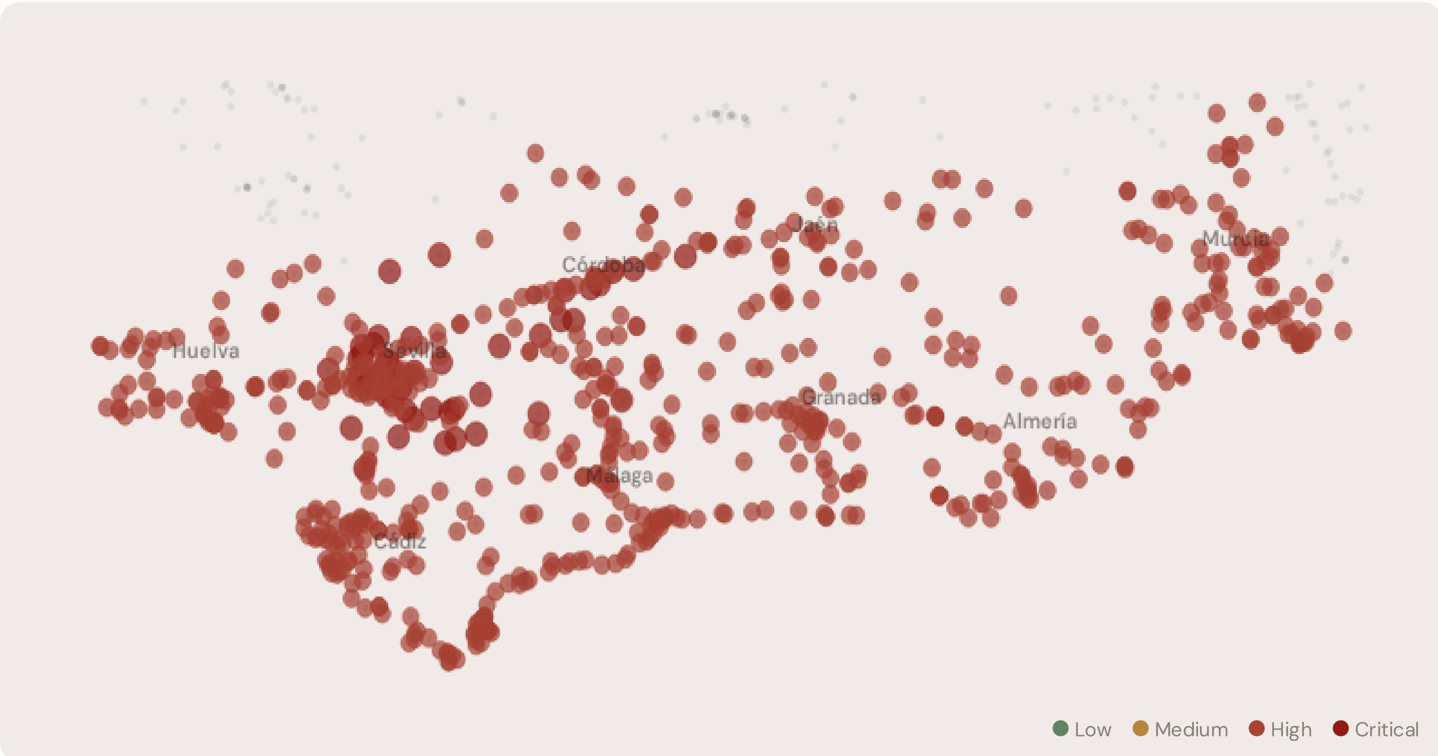
WORST PROVINCIA

Sevilla

Avg R: 0.742 · 122 substations

The Andalucía-Murcia hosts **688 substations** across six provincias, making it one of the most energy-critical corridors in Europe. However, its risk profile is disproportionately concentrated in the upper bands: **637 substations (92.6%)** are classified High and **51** Critical, with only **0** in the Low band. 60% of Andalucía-Murcia substations score above the national fleet median of 0.613. The corridor median R of 0.628 reflects the tension between seasonal demand volatility driven by tourism and agriculture, rapidly growing solar and wind capacity, and aging grid infrastructure not dimensioned for bidirectional renewable flows. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Spain 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación Las Cruces	Sevilla	0.798	Critical	0.430	0.397	0.450	0.183	0.801	0.826	S, T

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación Pintado	Sevilla	0.795	Critical	0.447	0.269	0.460	0.231	0.789	1.000	S, T
Subestación Morón de la Frontera	Sevilla	0.794	Critical	0.445	0.321	0.487	0.198	0.801	0.882	S, T
Subestación Islas	Sevilla	0.788	Critical	0.440	0.351	0.441	0.235	0.715	0.964	S, T
Subestación OSM 931784488	Sevilla	0.787	Critical	0.426	0.382	0.462	0.221	0.739	0.804	S, T
Subestación de Don Rodrigo	Sevilla	0.786	Critical	0.445	0.329	0.434	0.170	0.777	0.988	S, T
Subestación OSM 931776235	Córdoba	0.782	Critical	0.450	0.444	0.507	0.167	0.533	0.880	T
Subestación Giralt Laporta	Sevilla	0.778	Critical	0.428	0.298	0.441	0.205	0.770	0.914	S, T
Subestación Alcores	Sevilla	0.777	Critical	0.411	0.322	0.476	0.214	0.761	0.896	S, T
Subestación de Casaquemada	Sevilla	0.777	Critical	0.445	0.421	0.459	0.185	0.741	0.626	S

… 678 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — ANDALUCÍA-MURCIA VS FLEET

C — Continuity 0.429 vs 0.481 (−11%)

I — Infrastructure 0.518 vs 0.517 (0%)

S — Saturation 0.289 vs 0.319 (−9%)

V — Voltage 0.390 vs 0.512 (−24%)

E — Economic 0.215 vs 0.339 (−37%)

T — Transition 0.728 vs 0.663 (+10%)

MODIFIER AVERAGES — ANDALUCÍA-MURCIA VS FLEET

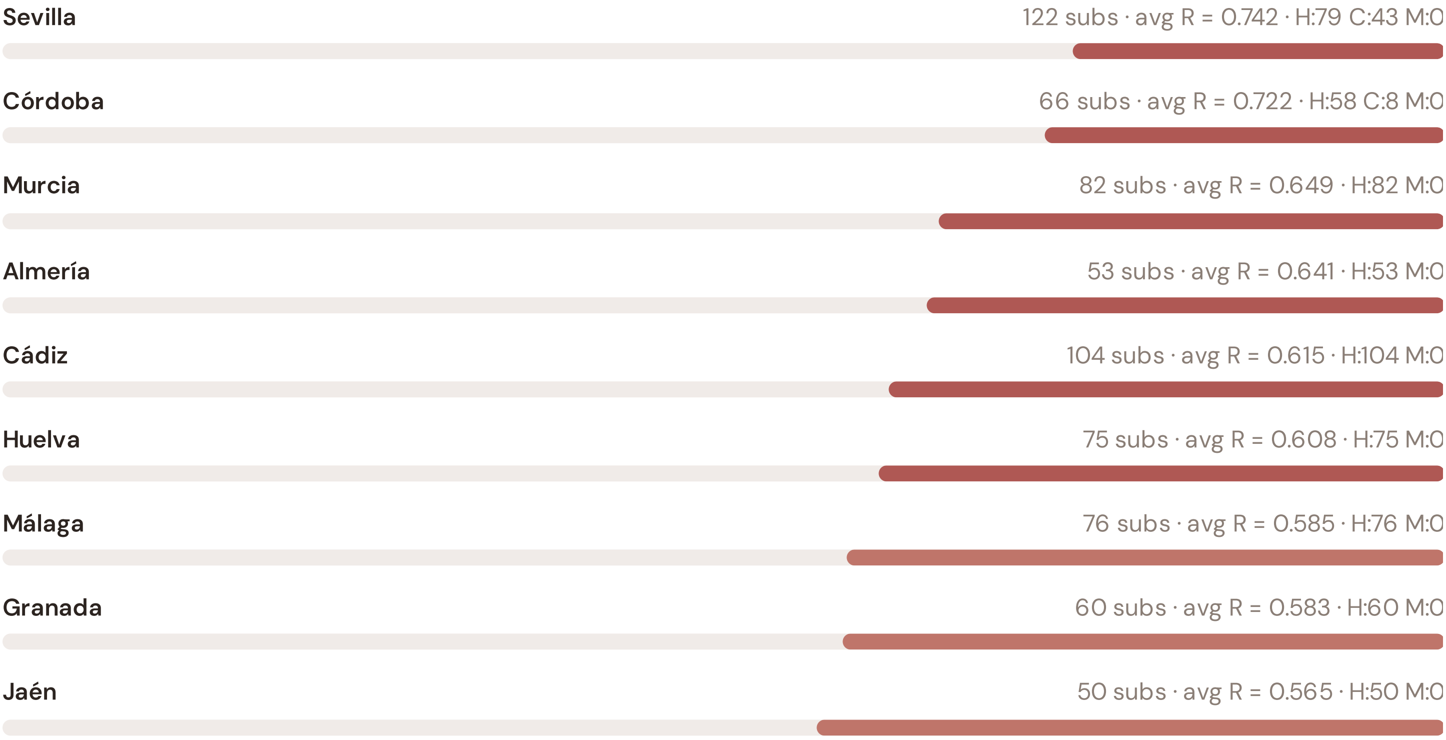
R3 Consequence	1.265	fleet 1.119	↑
R4 Graph Crit.	1.008	fleet 1.008	→
R6a Restoration	0.979	fleet 0.995	↓
R6b Seismic	1.249	fleet 1.200	↑
R7 Digital Read.	1.000	fleet 1.008	→

The dominant risk driver across the Andalucía-Murcia corridor is **T (Transition)**, averaging 0.728 against a fleet average of 0.663 — occupying 1457% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.390 vs fleet 0.512 (390% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.265 (fleet: 1.119), reflecting the socio-economic fragility of the Andalucía-Murcia corridor's agriculture-intensive, tourism-dependent economy with growing renewable capacity but limited grid interconnection. The R6b seismic modifier averages 1.249, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Provincia Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The province-level breakdown reveals significant intra-regional variation. **Sevilla** leads with a mean R of 0.742 across 122 substations, while **Jaén** scores 0.565 — a spread of 0.177 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Andalucía-Murcia is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

254 Andalucía-Murcia substations score R ≥ 0.650, placing them in the upper quartile of national risk. For PNIEC (Programmation Pluriannuelle de l'Énergie) investment prioritisation, this analysis identifies the Sevilla corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

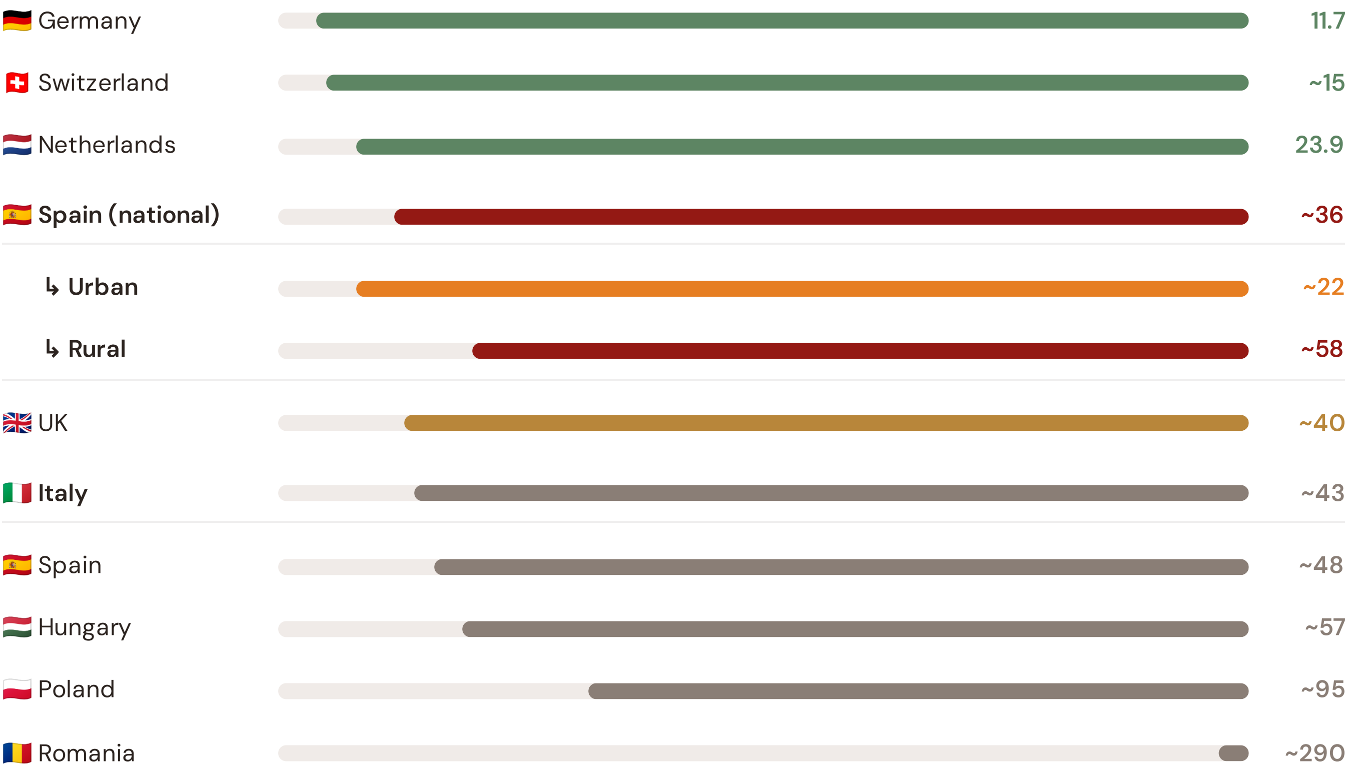
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Andalucía-Murcia corridor analysis hinges on Spain's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); CNMC TIQE (2023); Bundesnetzagentur (2024). Spain sub-categories from CNMC territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Andalucía-Murcia corridor deep-dive shows **688 substations** (of 688) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Spain's rural territory classification. The Andalucía-Murcia's average C of 0.429 is **0.9× the fleet average** of 0.481. In European terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~48–57 min/customer/year) than to Spain's own national average (~36 min). The SSI reveals what aggregate CEER statistics conceal: that Spain's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the Andalucía-Murcia corridor concentrates the worst of the latter.

Regional Risk Ranking — All 17 Comunidades Autónomas

Where does the Andalucía-Murcia sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1 Extremadura — 173 subs · avg R = 0.950 · H:0 C:173

#2 Castilla-La Mancha — 334 subs · avg R = 0.908 · H:0 C:334

#3 Aragón — 400 subs · avg R = 0.783 · H:60 C:340

#4 Canarias — 32 subs · avg R = 0.689 · H:32

#5 Castilla y León — 500 subs · avg R = 0.666 · H:436 C:64

#6 Región de Murcia — 82 subs · avg R = 0.649 · H:82

#7 Andalucía — 606 subs · avg R = 0.643 · H:555 C:51

#8 Illes Balears — 32 subs · avg R = 0.578 · H:32

#9 Comunitat Valenciana — 183 subs · avg R = 0.575 · H:183

#10 Galicia — 264 subs · avg R = 0.530 · H:201

#11 Comunidad Foral de Navarra — 176 subs · avg R = 0.441 · H:1

#12 Principado de Asturias — 81 subs · avg R = 0.398 · H:0

#13 Cantabria — 61 subs · avg R = 0.378 · H:0

#14 Cataluña — 301 subs · avg R = 0.367 · H:0

#15 La Rioja — 52 subs · avg R = 0.358 · H:0

#16 País Vasco — 104 subs · avg R = 0.257 · H:0

#17 Comunidad de Madrid — 148 subs · avg R = 0.240 · H:0

The Andalucía-Murcia ranks **#0 of 17 comunidades autónomas** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Extremadura, Castilla-La Mancha, Aragón, while the three most resilient are Comunidad de Madrid, País Vasco, La Rioja. 7 of 17 comunidades autónomas score above the fleet average of 0.614. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Spain. It is precisely this substation-level granularity that positions the SSI as a tool for PNIEC / España 2030 / Plan de Recuperación investment prioritisation: not treating entire comunidades autónomas as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the REE/DSO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	REE/DSO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a €10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against CNMC (Commission de Régulation de l'Énergie)'s published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Spanish training set comprises 3,529 substations with complete 95-feature vectors. Average CI_width of 0.265 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 153 Low, 832 Medium, 1582 High, 962 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The Extremadura Solar Frontier

Deep-dive into Extremadura's grid risk profile — substation map, component analysis, province breakdown. European Context Card: Massive PV build-out vs weak transmission capacity and rural grid topology.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

DGT are due for refresh in Q3. Impact: IDAE (Registro de Instalaciones de producción de energía eléctrica) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CNMC (Comisión Nacional de los Mercados y la Competencia)** (8 variables → C1–C4 (TIEPI/NIEPI/CAIDI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Extremadura**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Spanish utility, CNMC, REE, regional DSO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

Published 9 April 2026 · SSI Index v4.0.2 · 3,529 substations · ~22,765 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5